



East West University
Department of Computer Science and Engineering
Course Outline
Spring 2026 Semester

Course: CSE207 Data Structures

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	2 Hours/Week for 14 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

Prerequisite

CSE110 Object Oriented Programming

Instructor Information

Instructor: Dr. Maheen Islam, Associate Professor and Chairperson, CSE Dept.
Office: Room # 642
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TA: TBA

Class Routine and Office Hour

Course Objective

The course develops students' skills for designing and analyzing linear and non-linear data structures. It strengthens students' ability to identify and apply the suitable data structure for solving real-world problems. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE246 Algorithms, CSE366 Artificial Intelligence, CSE405 Computer Networks, and CSE 471 Compiler Design.

Knowledge Profile

K3: Theory-based engineering fundamentals

Learning Domains

Cognitive - C2: Understanding, C3: Applying, C4: Analyzing

Program Outcomes (POs)

PO1: Engineering Knowledge

PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required

EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	Interpret and Apply the basic concepts of linear lists for developing effective problem solutions.	PO1	C2, C3	K3	
CO2	Interpret and Apply the basic concepts of the non-linear list for manipulating hierarchical and connected data.	PO1	C2, C3	K3	
CO3	Choose and justify appropriate data structures for solving computational problems.	PO2	C2, C3	K3	
CO4	Analyze and Use the appropriate data structure and Write reports to design, build and test complex problems.	PO2	C4, A2, P2, P3	K3	EP1, EP2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Subtopics / Details		Course Topic	Teaching-Learning Method	CO	Class Tests			Mid-Semester Exam		Project	Final Exam		PO Marks	
					1	2	3							
					C2	C2	C2	C2	C3		C2	C3	PO1	PO2
Day 01	Familiarization on the course (Course Outline). Primitive and Non-Primitive data structures, Abstract Data Types (ADT) and its applications	Singly and doubly linked list Implementation and Basic operations with Application	Lecture, Class Discussion, Discussion Outside Class with Instructor/ Teaching Assistant	CO1, CO3	5			5	10				10	10
Day 02	Performance Analysis, Arrays, Time & Space Complexity													
Day 03	Pointers, Dynamic Memory Allocation, Classes & Objects													
Day 04	Linked Lists Introduction, Node Structure, Basic Operations													
Day 05	Singly Linked List: Deletion, Searching, Reversing													
Day 06	Doubly Linked Lists, Circular Linked Lists													
Day 07	Linked List Problem Solving													

Day 08	Stack: Definition, Array Implementation	Stack and Queue: Basic Operations and Application Implementation	Do	CO1 CO3				10	5				10	5
Day 09	Stack: Linked List Implementation													
Day 10	Stack Applications, Expression Conversion and Evaluation													
Day 11	Queue, Queue using Linked List, Applications													
Day 12	Circular Queue, Deque													
Day 13	Midterm Exam													
Day 14	Recursion, Problem Solving with Recursion	Different Trees: Basic operations and applications	Do	CO2 CO3							5	10	10	10
Day 15	Trees Introduction, Binary Trees, Traversals													
Day 16	Binary Search Tree, Insertion, Searching													
Day 17	BST Deletion, Advanced BST Operations													

Day 18	AVL Trees, Rotations, Balancing Operations														
Day 19	B-Trees, Red-Black Trees														
Day 20	Heaps, Priority Queue, Heap Sort														
Day 21	Graphs Introduction, Adjacency Matrix & List, Different properties of graph using List and matrix	Graph basic operations and implementation		CO2 CO3			5					5	5	10	5
Day 22	BFS: Implementation with List and Matrix														
Day 23	DFS: Implementation with List and Matrix														
Day 24	Graph Applications: Bipartite Graphs and others														
Day 25	Hashing, Collision Handling	Hashing: Hash table generation, Collision resolution		C02									5	5	
Day 26	Final Examination														

Laboratory Experiments and Assessment Scheme

Experiment	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels	Mark of Psycho-motor Learning Levels		Mark of Affective Learning Levels	Mark of COs
			C4	P2	P3	A2	
Implement program using pointers, structure and DMA etc.	Preparing Pre-Lab Report, Lab Experiment and Result Analysis, Preparing Post-Lab Report	CO4					
Implementation of different operations on singly linked list – copy, concatenate, split, reverse, count no. of nodes etc.	Do	CO4					
Implementation of different operations on doubly linked list – copy, concatenate, split, reverse, count no. of nodes etc.	Do	CO4					

Implementations of stack menu driven program.	Do	CO4					
Implementations of queue menu driven program.	Do	CO4					
Implementations of recursion.		CO4					
Implementations of the BST program.	Do	CO4					
Implementations of the AVL tree program.	Do	CO4					
Implementations of graphs using arrays and linked lists	Do	CO4					
BFS & DFS and problem-solving using graph	Do	CO4					
Lab Experiments		CO4	2	1	1	1	5
Lab test	Individual Lab Test		5	2	2	1	10

Total			7	3	3	2	15
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Mini Projects

Mini Project	Teaching-Learning Method	CO	EP/ EA	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
				C4	P2	P3	A2	
Mini Project including Report and Presentation	Group-based, moderately complex electrical circuit building for practical application with report writing and presentation	CO4	EP1, EP2	2	1	1	1	5

Overall Assessment Scheme

Assessment Area	CO	Assessment Area Mark
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	CO1	CO2	CO3	CO4	
Attendance					5
Class Test/Quiz	5	5	5		15
Mid Semester Assessment	20	0	10	0	30
Final Exam	0	20	10	0	30
Laboratory Performance and Lab VIVA	0	0	0	5	5
Lab Final				10	10
Mini Project	0	0	0	5	5
Total	25	25	25	20	100

Teaching Materials/Equipment

Textbook

- Gilberg, Richard, and BehrouzForouzan. Data Structures: A pseudocode approach with C, 2nd Edition, Publisher:Nelson Education, 2004.

References

- Aho, Alfred V., and Jeffrey D. Ullman. Data structures and algorithms. Publisher: Pearson, 1983
- Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. Introduction to algorithms. Publisher:MIT press, 2009

Lab Manual:

Lab manual will be provided.

Project Description:

Project description will be provided.

Equipment/Software:

Any C/C++ IDE: As example, Visual C++, Code::Block, and/or Dev-C++

Exam Dates

As per the schedule provided by the department and the university.

Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00	45% to less than 50%	C	2.25
75% to less than 80%	A	3.75	40% to less than 45%	D	2.00
70% to less than 75%	A-	3.50	Less than 40%	F	0.00
65% to less than 70%	B+	3.25			
60% to less than 65%	B	3.00			
55% to less than 60%	B-	2.75			

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, or falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offense under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, the student **MUST** get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application.
- For **final exam**, there will be **NO** makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones **MUST** be turned to silent mode during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**